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UNDERLYING DEEP LEARNING NETWORKS DIAGNOSIS EVALUATION AND GENERATIVE ADVERSARIAL NETWORK DATA AUGMENTATION BASED ON A BENCHMARK ACCIDENT DATASET

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ABSTRACT

Nuclear energy plays an important role in the global energy supply, particularly as a key source of low-carbon electricity, while ensuring the safe operation of nuclear power plants (NPPs) is crucial. Given the significant impact of human-caused errors on three serious nuclear accidents in history, artificial intelligence (AI) has been increasingly interested assisting operators in NPP accident diagnosis. However, the field has lacked a benchmarking effort for underlying deep learning networks based on public datasets for accident diagnosis, as well as research on time-series data augmentation efforts for accident data. Based on the first open-source NPP accident dataset NPPAD, which covers all kinds of accident data of pressurized water reactors and has been validated with Fukushima nuclear accident data for reliability, this study explored accident diagnostic techniques based on Multilayer Perceptions (MLPs), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs) and Transformers, and data augmentation techniques based on conditional Generative Adversarial Networks (cGANs) and Time-series Generative Adversarial Networks (TimeGANs). Experimental results show Transformers have the best classification ability for accident types. In addition, the standardization of data has a large impact on the classification effect, and standardization in data preprocessing will make all models substantially more effective. For data augmentation, TimeGANs exhibit higher performance compared to cGANs in generating fake datasets and can better capture the dynamic features of time-series data. This paper provides a reference for the application of deep learning through the study of three parts: a comprehensive comparison of underlying deep learning networks, data standardization processing, and data augmentation.

Keywords: Accident diagnosis; Data Augmentation; Deep learning networks; Generative adversarial networks

1. INTRODUCTION

Nuclear energy has been controversial since its inception, because nuclear energy brings efficient and clean energy, it also brings the risk of nuclear accidents. Obviously, nuclear accidents cannot be eliminated, and any new system design or measures can only reduce the probability of accidents, which must be adequately prepared for both on-site and off-site emergencies. Experience from the Fukushima nuclear accident has shown that rapid diagnosis of nuclear power plant accidents after an accident is critical, as accident diagnosis is the primary aspect of nuclear emergency. However, NPPs contain a massive number of sensor parameters, and the ability of humans to quickly determine the type of accident from these parameters is on the weak side, and AI shows great potential in this regard. AI presents significant advantages in various critical aspects. First, AI demonstrates superior efficiency in information processing. Second, AI offers a notable reduction in human error. Third, AI possesses an inherent capability for continuous operation. As a typical representative of AI, deep learning (DL) techniques are the basis for numerous studies proposing methods to diagnose accidents in NPPs. The underlying deep learning networks include MLPs, CNNs, RNNs, Transformers. Hadad et al. worked on classifying design basis accidents using MLPs[1][2]. Liu et al. studied NPP accident diagnosis methods based on CNNs[3][4][5]. Choi et al. proposed NPP diagnosis methods based on RNNs and its variants[6][7][8][9]. Yi S et al. used Transformers for NPP accident diagnosis[10]. Qi et al. In summary, extensive research has been conducted within the community on underlying deep learning networks and their variants or combinations for accident diagnosis in nuclear power plants. Qi et al. have carried out research on various methods for combining networks, such as MLP-BN, LSTM-CNN, CNN-LSTM-MLP[11][12][13]. The above studies demonstrated the feasibility of underlying deep learning networks for NPP accident diagnosis but lacked a

comprehensive comparison based on a benchmark dataset. We have previously developed the first reliability-validated open-source dataset of nuclear power plant accidents, NPPAD[14][15]. In this paper, we carry out comparative network performance tests based on NPPAD, and provide comparative results under the same benchmark dataset to the nuclear field. To the best of our knowledge, we have also conducted the first research on enhancement techniques for NPP accident time-series datasets.

2. METHODS

2.1 Accident Diagnosis: Underlying Deep Learning Networks

Many studies proposed NPP accident diagnosis methods based on underlying deep learning networks, which mainly include MLPs[16], CNNs[17], RNNs[18] and Transformers[19]. As shown in Fig. 1, MLPs are characterized by an input layer that accepts any type of input, for instance, features from accidents, several possible interconnected so-called hidden layers, and an output layer with the predicted value, such as an accident type. CNNs use convolution operations or use what is called linear spatial operation, in at least one of their layers. The convolution layers learn filters that detect features, or patterns, in the input signals. The filters automatically capture features in a local region called the receptive field, and to capture features more distantly, CNNs learn a hierarchy of features. The pooling operation after the convolution layer reduces the dimensionality of the captured features. CNNs have been very successful when analyzing image and audio input data, but have also been used to analyze sequence data (for instance, Time-series data). ResNet and DenseNet are two common variants of CNNs. RNNs have gained much attention since they were first introduced and have been applied widely in many Natural Language Processing tasks, such as speech recognition. They are suitable for modeling sequential data such as text or time-series data. The elements in the sequence, for example, accident features, are processed step-by-step one at a time using so-called recurrent connection units, where the output of each step depends on both the current and the previous steps. Common RNNs include the long short-term memory (LSTM) model and the gated recurrent units (GRUs).

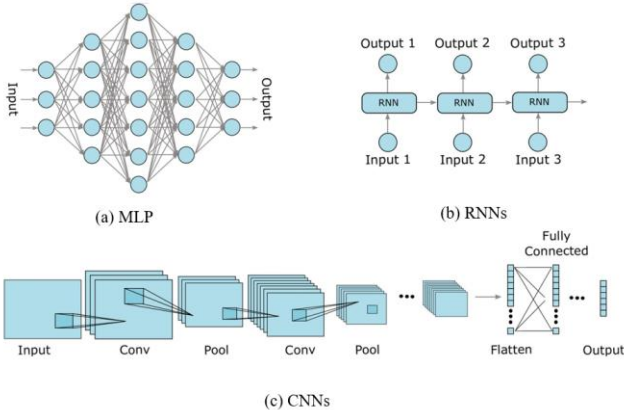


FIGURE 1: THE STRUCTURES OF MLP, CNNs AND CNNs.

As shown in Fig.2, the Transformer model was introduced in 2017 by Vaswani et al. and achieved state-of-the-art results in language translation using only a fraction of the previous training times. It is an encoder-decoder type of model, where the encoder maps the vector representations of the tokens from an input text (the input embeddings) to an internal representation. The decoder then uses the internal representation and maps it to the output sequences (for instance, the target language). Compared with similar models of the time, instead of using recursive layers, or convolutional layers, Transformers used an architectural component called attention. The attention module enables a model to consider the interactions among every pair of tokens in a sequence and automatically learn the relationships between tokens in a sequence that are relevant to the task at hand.

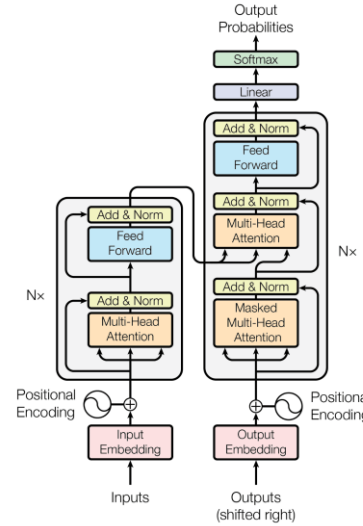


FIGURE 2: THE STRUCTURE OF TRANSFORMERS.

2.2 Data Augmentation: TimeGANs and cGANs

Due to the high safety and reliability of nuclear power plants, few nuclear accidents have occurred since the birth of commercial NPPs, and currently, only limited-sized datasets can be constructed using simulation data. Whereas GANs, as a generative model, can generate fake data, we would like to augment the original dataset by generating more data. Therefore, cGANs[20] and TimeGANs[21] as two typical variants of GANs are used for data augmentation studies.

TimeGANs were a time-series data generation model proposed by Jinsung Yoon et al. in NeurIPS 2019. The main idea is to combine the versatility of unsupervised GAN methods with the principle of conditional probability provided by supervised autoregressive models. TimeGANs can generate time series that retain time dynamics. As shown in Fig.3, TimeGANs consist of four network components: embedding function, recovery function, sequence generator, and sequence discriminator.

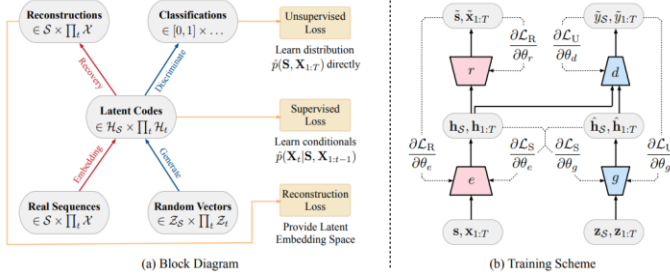


FIGURE 3: TimeGANs, (a) BLOCK DIAGRAM OF COMPONENT FUNCTIONS AND OBJECTIVES. (b) TRAINING SCHEME; SOLID LINES INDICATE FORWARD PROPAGATION OF DATA, AND DASHED LINES INDICATE BACKPROPAGATION OF GRADIENTS.

The original GANs cannot generate data according to the specified labels (for instance, accident types). cGANs (Fig. 4) were proposed by Mehdi Mirza et al, which are to incorporate the labeling information y into the generator G and the discriminator D . The cGANs can be used to generate accident data according to the specified labels.

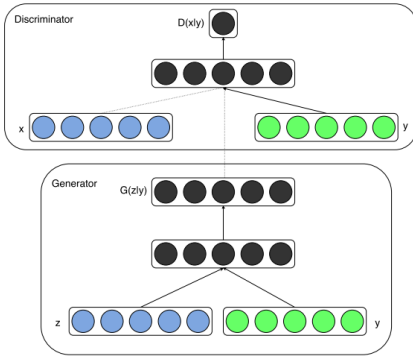


FIGURE 4: cGANs.

3. EXPERIMENTS

3.1 Dataset Description

The dataset used in this paper comes from previous work[14]. As shown in Fig. 5, we presented a first-of-its-kind open dataset created using PCTTRAN, a pre-developed and widely used simulator for NPPs. The dataset, namely nuclear power plant accident data, basically covered the common types of accidents in typical pressurized water reactor NPPs and contained time-series data on the status or actions of various subsystems, accident types, and severity information. Moreover, the dataset incorporates other simulation data (e.g., radionuclide data) for conducting research beyond accident diagnosis.

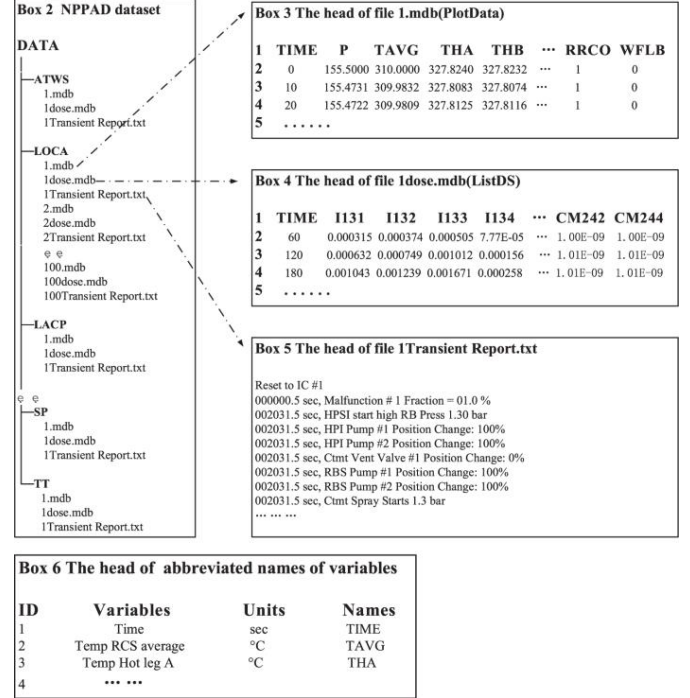


FIGURE 5: NPPAD, BOX 2 ILLUSTRATES THE GENERAL STRUCTURE OF THE DATA RECORDS. BOX 3 REPRESENTS THE TIME SERIES OF THE STATUS PARAMETERS WITH A 1% OF 100 CM2 BREAK OF LOCA. BOX 4 REPRESENTS THE TIME SERIES OF THE RADIONUCLIDE IN THE NPPS. BOX 5 DESCRIBES THE ACTIONS IN THE SUBSYSTEMS OF THE NUCLEAR PLANT OVER THE SIMULATION TIME FOR EACH ACCIDENT IN ADDITION TO THE MDB FORMAT MOREOVER. BOX 6 DESCRIBES THE PARAMETERS CORRESPONDING TO THE ABBREVIATIONS IN PLOTDATA.

3.2 Data Preprocess

The original dataset was shown in Table 1, which belongs to the balanced distribution dataset with 96 accident features. In this paper, we intercepted the data of 960s after the occurrence of each accident (sampling interval of 10s), and got each accident data with dimensions of [96, 96]. Due to the large order of magnitude difference between the features, normalization as shown in Eq. (1) was used.

$$x^* = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad (1)$$

where x is the original value, $\min(x)$ is the feature minimum and $\max(x)$ is the feature maximum.

TABLE 1: THE ACCIDENT INFORMATION OF NPPAD.

Accident	Description	Number	Severity
LOCA	Loss of Coolant Accident (Hot Leg)	100	% of 100 cm ²
LOCAC	Loss of Coolant Accident (Cold Leg)	100	% of 100 cm ²
SLBIC	Steam Line Break Inside Containment	101	% of 100 cm ²
SLBOC	Steam Line Break Outside Containment	100	% of 100 cm ²
SGATR	Steam Generator A Tube Rupture	100	% of 1 full tube rupture
SGBTR	Steam Generator B Tube Rupture	110	% of 1 full tube rupture
RW	Rod Withdrawal	100	% (+/-) withdrawn
RI	Rod Insertion	82	% (+/-) insertion
FLB	Feedwater Line Break	100	% of 100 cm ²
MD	Moderator Dilution	100	% of unbored injection
LR	Load Rejection	99	% of full load rejected
LLB	Letdown Line Break in auxiliary buildings	101	% of nominal letdown flow

3.3 Underlying Networks Performance Experiments

This section conducted performance experiments on the underlying deep learning network based on NPPAD, where 80% of the dataset was used for training, 10% for validation, and 10% for testing. To fully test the performance of these networks, we used multiple variants of them. For example, ResNet and DenseNet were introduced for CNNs. In addition, sourced a chance experiment, we experimented with using Batch Normalization (BN)[22] at the first layer for all models and found that it provided performance improvements for all models. This was the reason why the BN layer is considered for each model experiment in the following.

(1) MLPs

Four models were selected for the MLPs: a one-layer fully connected layer and a one-layer fully connected layer with a first BN layer, a four-layer fully connected layer, and a four-layer fully connected layer with a first BN layer for the MLP network, and all four models were connected to a *Softmax* activation function as a classifier. The loss curves of these four models are shown in Fig. 6, and the confusion matrix for the best models is in Fig. 7.

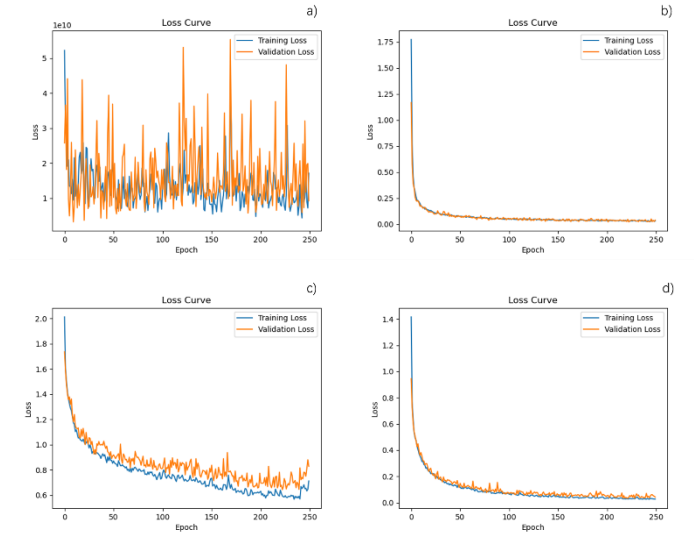


FIGURE 6: LOSS CURVES FOR TRAINING AND VALIDATION DURING MLP TRAINING, (a) MLP1 (b) MLP1(BN) (c) MLP4 (d) MLP4(BN).

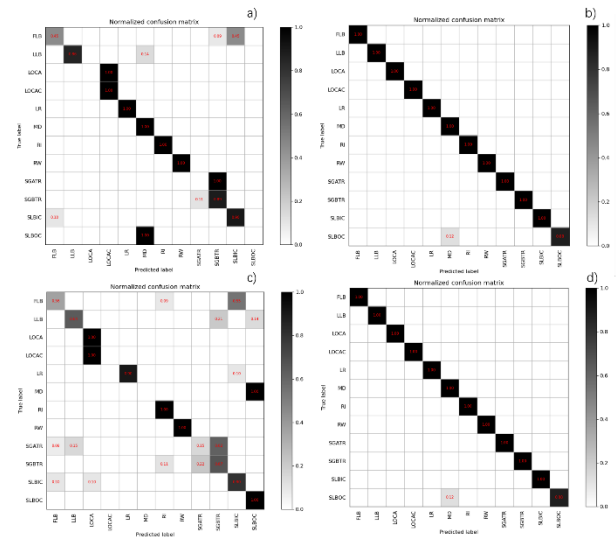


FIGURE 7: CONFUSION MATRIX FOR THE BEST MODELS, (a) MLP1 (b) MLP1(BN) (c) MLP4 (d) MLP4(BN).

(2) CNNs

The model mainly used shallow ResNet and DenseNet, with ResNet_3 indicating that a 3-layer ResNet BasicBlock was used and DenseNet_4 indicating that a 4-layer DenseBlock was used. The loss curves of these four models are shown in Fig. 8, and the confusion matrix for the best models is in Fig. 9.

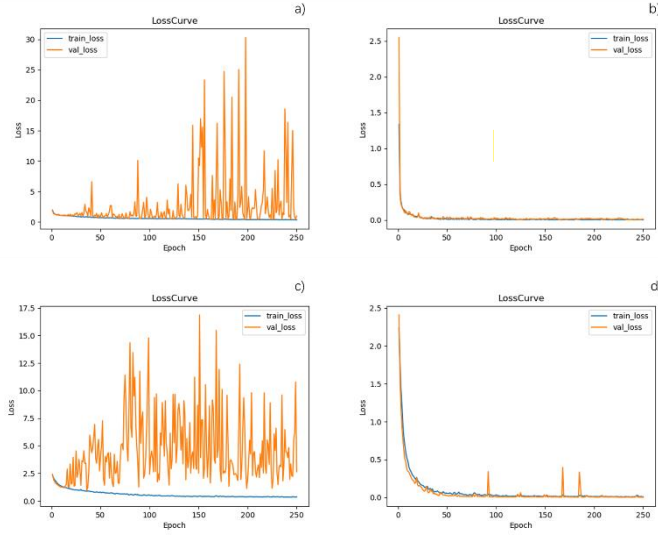


FIGURE 8: LOSS CURVES FOR TRAINING AND VALIDATION DURING CNN TRAINING, (a) ResNet (b) ResNet (BN) (c) DenseNet (d) DenseNet (BN).

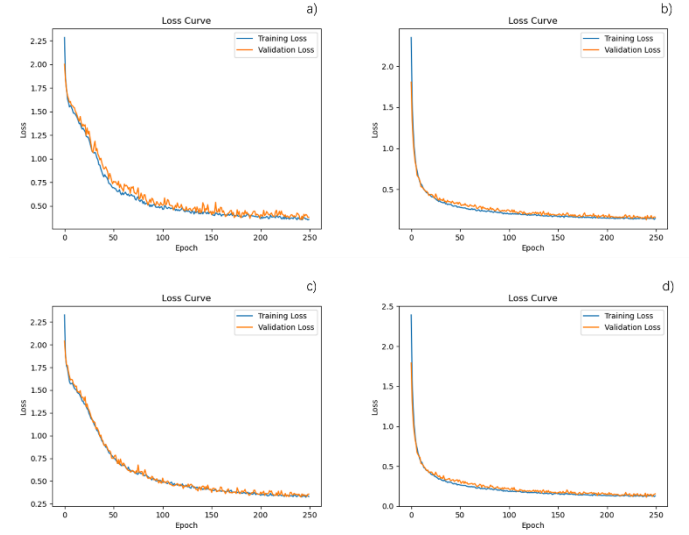


FIGURE 10: LOSS CURVES FOR TRAINING AND VALIDATION DURING RNN TRAINING, (a) GRU (b) GRU(BN) (c) LSTM (d) LSTM(BN).

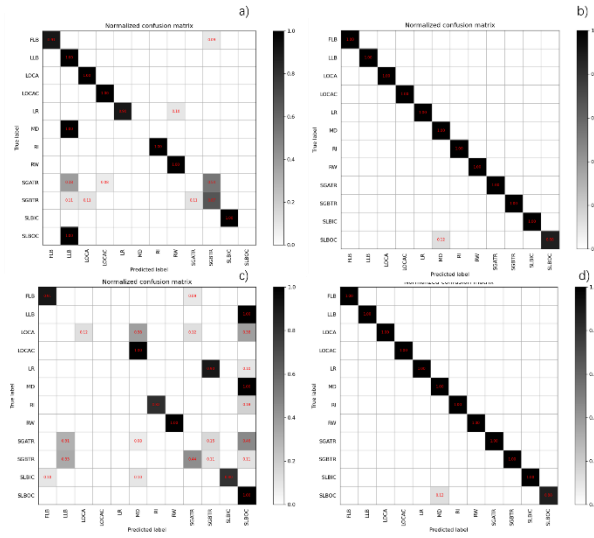


FIGURE 9: CONFUSION MATRIX FOR THE BEST MODELS, (a) ResNet (b) ResNet (BN) (c) DenseNet (d) DenseNet (BN).

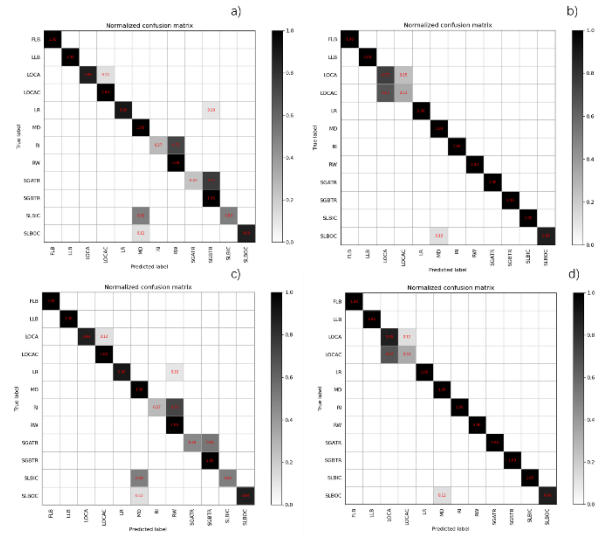


FIGURE 11: CONFUSION MATRIX FOR THE BEST MODELS, (a) GRU (b) GRU(BN) (c) LSTM (d) LSTM(BN).

(3) RNNs

Four models were also selected for RNNs: GRU network with two GRUCells and GRU network with two GRUCells with first BN layer, LSTM network with two LSTMCells and LSTM network with two LSTMCells. Four models were connected to the first BN layer, the fully connected layer, and the *Softmax* activation function as a classifier. The loss curves of these four models were shown in Fig. 10, and the confusion matrix for the best models is in Fig. 11.

(4) Transformers

Two models were selected for Transformers: Transformers-encoder and Transformers-encoder with BN layer. These two models employed a Transformer consisting of an Encoder module, with the final addition of a fully connected layer and average pooling outputting the probabilities of the 12 categories. The loss curves of these two models were shown in Fig. 12, and the confusion matrix for the best models is in Fig. 13.

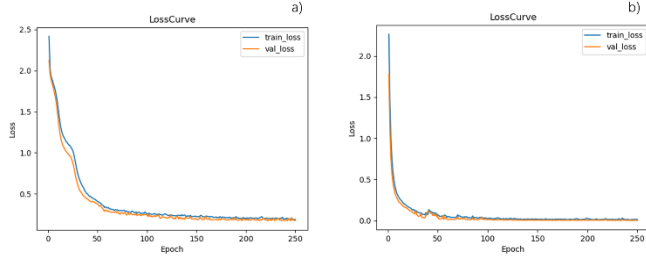


FIGURE 12: LOSS CURVES FOR TRAINING AND VALIDATION DURING TRANSFORMERS TRAINING, (a)Transformers-encoder (b) Transformers-encoder (BN).

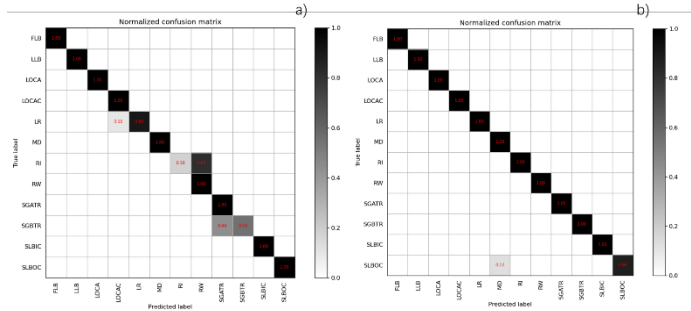


FIGURE 13: CONFUSION MATRIX FOR THE BEST MODELS, (a)Transformers-encoder (b) Transformers-encoder (BN).

(5) Comprehensive comparison

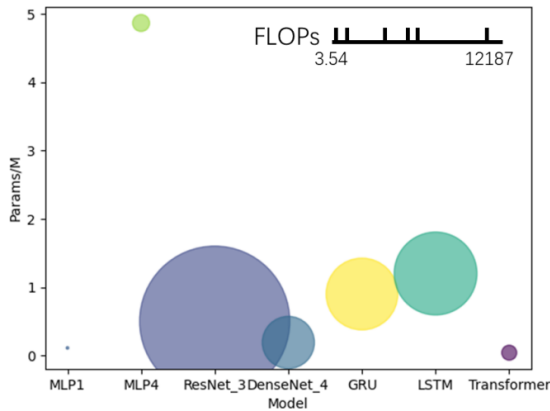


FIGURE 14: TWO-DIMENSIONAL PLOT OF MODEL FLOATING-POINT COMPUTATION SPEED AND NUMBER OF PARAMETERS.

TABLE 2: COMPREHENSIVE COMPARISON OF THE PERFORMANCE OF ALL MODELS.

Models	FLOPs	Params	FLOPs/Params	Accuracy
MLP1	3.54M	0.11M	32.18	67.23%
MLP1(with first BN layer)				99.16%
MLP4	155.84M	4.87M	32.00	62.18%
MLP4(with first BN layer)				99.16%
ResNet_3	12187.01M	0.50M	24372.02	67.23%
ResNet_3 (with first BN layer)				99.16%
DenseNet_4	1469.34M	0.19M	7733.37	38.66%
DenseNet_4 (with first BN layer)				99.16%
GRU	2777.86M	0.90M	3086.51	78.15%
GRU(with first BN layer)				95.80%
LSTM	3701.30M	1.20M	3084.42	80.67%
LSTM(with first BN layer)				96.64%
Transformer	121.50M	0.04M	3037.5	88.24%
Transformer(with first BN layer)				99.16%

The number of model floating-point computations was in descending order: MLP1 << Transformer < MLP4 << DenseNet_4 < GRU < LSTM << ResNet_3. The number of parameters was in descending order: Transformer < MLP1 < DenseNet_4 < ResNet_3 < GRU < LSTM < MLP4. There was consistency in the relative sizes of the model parameter sizes and the number of floating-point computations for the same networks, but there was no consistency for networks that were not of the same type. This was because the number of floating-point computations corresponding to the unit parameters (FLOPs/Params) were: MLP1 $\approx$ MLP4 << Transformer $\approx$ LSTM $\approx$ GRU < DenseNet_4 < ResNet_3. FLOPs/Params reflected the time cost required at the cost of unit space for different models, and FLOPs/Params were relatively consistent across models. The fully connected layer of MLP belonged to the linear model, which can be simply expressed as $WX+B$, and a pending parameter only needed to be executed once when there was no batch processing, and since the size of the batch selected was 32, the FLOPs/Params was approximately equal to 32. Compared with MLPs, RNNs, and Transformers had an additional timing processing part, and the number of computations required for a single parameter to be processed was $\text{batch_size} \times \text{sequence length} = 32 \times 96 = 3072$ in the case of parameter sharing. CNNs involved rolling products of convolutions (related to the size of the convolution kernel) and direct connections, where the situation was more complex. However, it was certain that the FLOPs/Params of CNNs are slightly larger than those of RNNs and Transformers, and much larger than those of MLPs.

(6) DISCUSSION

(a) Models

Comparing the accuracy of several models, MLP1(BN) = MLP4(BN) = ResNet_3(BN) = DenseNet_4(BN) = Transformer (BN) > LSTM(BN) > GRU(BN) > Transformer > LSTM > GRU > ResNet_3 > MLP1 > MLP4. When the model didn't have the first BN layer, the best results were obtained with Transformers, followed by RNNs (LSTM and GRU). This reflected that the temporal information of the data plays a more

important role in determining the accident type. It was noteworthy that Transformers not only had the highest accuracy but also had the smallest number of parameters and computational effort. The result of CNNs was inferior to the above two types of models, and each case of accident data (96*96 size) was treated as an image in CNNs. However, the accident data didn't have two-dimensional structural information like a natural image, so the effect was average. The accuracy of MLP had little relationship with the number of hidden layers, the model becomes bigger, and the effect may not become better. In addition, adding a nonlinear activation function to the MLP layer with fewer layers reduced the accuracy of the model. It was found that the ReLU function affects the gradient update when the number of network layers was too small, and the effect is well improved by using the Sigmoid function instead of the ReLU function. The effect was further improved by removing the activation function, which was due to the presence of some linearly differentiable features in the data. The reason was that there were some linearly differentiable features in the data, and when there were fewer hidden layers, adding a nonlinear activation function affected their linear differentiability.

(b) Data preprocessing

In our experiments, it was found that after adding a BN layer in front of the model, the accuracy of the model was improved to more than 95%, and even a layer of MLP can classify accidents very accurately. It is because the BN layer acted similarly to normalization (subtracting the mean, dividing by the variance) in data preprocessing, with the difference being only that the BN layer uses the mean and standard deviation of the batch in training, whereas data normalization used the mean and standard deviation of the entire data, and the BN layer contains two learnable parameters (scale parameter γ and shift parameter β). Outputting γ and β for the BN layer of the trained model, $\gamma \approx 1$ and $\beta \approx 0$, and it can be assumed that this BN layer serves approximately the same effect as data normalization. We conducted experimental validation by removing the first BN layer of the model and adding the normalization operation to the data preprocessing and found that the final accuracies were all above 95% as well, validating our idea.

At first, we only normalized the dataset, and the effect was much different compared to the later standardization. We believe that the reason is that there are more outliers in the accident data and they are far away from the mean, which makes the normalization ineffective, while the standardization avoids this problem and maintains the spacing of the samples better.

(c) Confusion Matrix

According to the confusion matrix of each model, it can be found that FLB and RW belong to the types that can be identified more accurately in all models, while SLBOC and MD were easy to confuse, and SLBOC had a part of the data that can be easily misclassified as MD in almost most of the models. Except for SLBOC and MD, most of the other categories can be better recognized. Preprocessing can be categorized better.

3.4 Data Augmentation Experiments

Due to the high safety and reliability of nuclear power plants, very few nuclear accidents have occurred since the birth of commercial nuclear power plants and currently only limited-sized datasets can be constructed using simulation data. Whereas GAN, as a generative model, can generate fake data, we would like to augment the original dataset by generating more data with the help of the GAN model. Therefore, this section explored the possibility of generating nuclear power plant accident data based on cGANs and TimeGANs. The performance of the two models was compared by the following two discriminatory criteria:

(1) Fidelity (FID): since the time series data has distinct physical characteristics, inspired by the idea of image generation in the field of vision, it is possible to convert the generated data and the real data into images with physical characteristics. Therefore, the performance of the two models is compared by calculating the FID values of the two images.

(2) Accident diagnosis score: the accident diagnosis model of NPPs is trained with the generated data and tested on the real data set, and the accuracy of the real data test set is used as the score.

TABLE 3: COMPREHENSIVE COMPARISON OF THE PERFORMANCE OF ALL MODELS.

Models	Generated data (cGAN)	Generated data (TimeGAN)	Real data	Mixed data (Real+cGAN)	Mixed data (Real+TimeGAN)
MLP1	19.33%	6.72%	67.23%	94.12%	97.48%
MLP2	5.04%	5.88%	62.18%	93.28%	96.64%
Resnet	10.92%	19.33%	67.23%	100.00%	99.16%
DenseNet	10.92%	4.20%	38.66%	99.16%	99.16%
GRU	0.84%	26.05%	78.15%	86.55%	92.44%
LSTM	0.84%	17.65%	80.67%	89.92%	93.20%
Transformer	5.88%	13.45%	88.24%	99.16%	99.16%

The fidelity comparison was shown in Tab. 4(1) and Tab. 4(2), and the quality of TimeGANs generated data is significantly higher than that of cGANs by comparing the image's physical characteristics and FID value. The accident diagnosis score was shown in Table 3, and the dataset was generated using the two models separately according to the ratio of 1:1 with the real dataset. Five datasets were constructed, and two mixed datasets were composed of 80% real and generated data. The same test set (10% real dataset) was used for all five datasets, with higher accuracy proving the higher quality of the model-generated data. Overall, the performance of the models trained on the generated data was average due to the small sample size of the real dataset, which makes the training of the generated models an inadequate fit. However, compared to real data, the performance of models trained on mixed data was effectively improved. Comparing the performance of the two models individually, the accuracy of TimeGANs was higher than that of cGANs in most of the cases, apparently because TimeGANs can better capture the temporal dynamics of the temporal dataset.

TABLE4(1): COMPARISON OF FIDELITY.

Accidents	Real data	Generate data (cGAN)	Generate data (TimeGAN)	FID (cGAN)	FID (TimeGAN)
FLB				549.2	302.2
LLB				510.1	318
LOCA				498.5	191.2
LOCAC				489.4	174.3
LR				602.7	377.5
MD				509.0	310.2

TABLE4(2): COMPARISON OF FIDELITY.

Accidents	Real data	Generate data (cGAN)	Generate data (TimeGAN)	FID (cGAN)	FID (TimeGAN)
RI				525.8	233.9
RW				545.6	284.8
SGATR				521.5	332.9
SGBTR				514.9	305.0
SLBIC				531.2	282.1
SLBOC				467.0	250.7

3.5 Summary

This part first studied the accident diagnosis performance of underlying deep learning networks, including MLPs, CNNs, RNNs, and Transformers. First, Transformers had the best

comprehensive performance, as evidenced by the highest diagnostic accuracy and the smallest number of model parameters and operations. Second, data preprocessing had a large impact on model diagnostic accuracy, and the introduction of the BN layer for all models resulted in a large improvement in model diagnosis accuracy, and the MLP accuracy reached the highest. Through the validation experiments, it was found that standardizing the data uniformly achieved almost the same effect as the BN layer, which was because the accident data had more outliers and deviated farther from the mean, which led to poor normalization. Standardization can avoid this problem and keep the spacing of samples better. Then data augmentation techniques were explored, and two generative models were introduced, cGANs and TimeGANs, for fake data generation. It was found that both can improve the performance of all models after introducing the fake dataset, and TimeGANs show stronger performance in data augmentation due to better capture ability of the temporal features.

4. CONCLUSION

This paper studies the comprehensive comparison of NPP accident diagnosis models based on underlying deep learning networks and data augmentation techniques based on cGANs and TimeGANs. It can be concluded that:

(1) Data preprocessing impact: the methods have a large impact on the NPP accident diagnosis accuracy for all deep learning networks. The standardized dataset gives a more satisfactory performance even on the MLPs with the activation function removed.

(2) Superiority of TimeGANs in data augmentation: TimeGANs show higher performance compared to cGANs, and TimeGANs can better capture the dynamic features of time-series data.

(3) Benchmark dataset performance: the results show that Transformers have the best classification ability for NPP accidents.

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